Same function, different pruning: the dominant direction of $\boldsymbol{W}_O$ under free, soft, and hard intervention

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Abstract

A widespread practice in attention-head analysis reads the geometry of the output projection $\boldsymbol{W}_O$: the first singular vector $\boldsymbol{v}_1(\boldsymbol{W}_O^{(h)})$ of each head is taken as its representative direction for pruning, interpretation, or routing, under the assumption that this geometry reflects its function. This work tests that assumption at three intensities of intervention on the value-output sector: under a free gauge the direction drifts with no functional effect; under a probe that separates it up to the simplex threshold, neither; only the hard imposition costs function. The coupling is one-directional, and it has a closed-form explanation: $\boldsymbol{v}_1(\boldsymbol{W}_O)$ is not identifiable under a reparametrization of the value-output sector that does not touch the function, while the OV circuit $\boldsymbol{W}_v\boldsymbol{W}_O$ —the sector's only identifiable object— remains fixed. The consequence is an impossibility: no similarity threshold on that direction admits a positive certified radius, and the pruning decision it produces changes under gauge in more than 90 % of cases, both on a vision-finetuned ViT and, by the same closed form and without any training, on a pretrained language transformer. As an instrument, the gauge-invariant response signature $\boldsymbol{v}_1(\boldsymbol{C}_h^P)$ measures functional diversity without gauge ambiguity, and is head-specific. Results are established on a ViT-B/16 finetuned on ImageNet-100 with five seeds and contrasted on a ViT-L/16 and a frozen DINOv2: non-identifiability and signature specificity reproduce across the three columns, and inertness across the two finetuned ones, while the relocation toward $Q\cdot K$ —correlational, with a reproducible crossover at layer 5— is bounded to the finetuned supervised regime.

Keywords: multi-head attention, gauge non-identifiability, certified impossibility, OV circuit, functional diversity, head pruning, Vision Transformers.

1 Introduction

The Vision Transformer has displaced the convolutional network as the reference architecture for image classification, and its multi-head attention mechanism is presented as a way for the model to attend, in parallel, to distinct spatial relations. Empirical evidence qualifies that reading: a fraction of the heads converge on nearly indistinguishable attention patterns, so that the layer's nominal capacity does not translate into effective diversity. Pruning heads after training barely degrades accuracy [8, 9], indicating that the redundancy is not marginal but structural.

On top of that observation a practice has been built that deserves to be made explicit. To decide which heads are redundant —and prune or merge them—, to interpret the role of a head, or to route information between them, a family of methods reads the *geometry of the output projection* $\boldsymbol{W}_O$: the first singular vector $\boldsymbol{v}_1(\boldsymbol{W}_O^{(h)})$ of each head is taken as its representative direction, under the assumption that this geometry reflects what the head does. The assumption is testable, and this work tests it. Its answer is negative: the dominant direction on which that practice operates does not carry the function of the head.

The argument rests on three intensities of intervention on that geometry, read together. Under the *free gauge* —reparametrizing $\boldsymbol{W}_v \to \boldsymbol{W}_v \boldsymbol{R}$, $\boldsymbol{W}_O \to \boldsymbol{R}^{-1} \boldsymbol{W}_O$ with $\boldsymbol{R}$ invertible leaves the function intact— $\boldsymbol{v}_1(\boldsymbol{W}_O)$ drifts at will while the OV circuit $\boldsymbol{W}_v \boldsymbol{W}_O$, the sector's only identifiable object, remains fixed: reading that direction is reading a choice of gauge. Under the *soft probe*, which angularly separates the representative directions up to the simplex threshold, the geometry moves tens of degrees without the functional similarity between heads separating from the between-seed noise. Under the *hard variant*, which imposes that same simplex by construction, the function does suffer. The coupling is one-directional: the geometry of $\boldsymbol{W}_O$ does not govern the function except when it is forced until it breaks, and that asymmetry —not non-identifiability by itself— is what the evidence supports. On top of it, a layer-wise characterization further places the organization of attention in the $Q \cdot K$ interaction, toward which it relocates with depth in the finetuned regime, oblivious to the dominant direction of $\boldsymbol{W}_O$.

If the dominant direction does not carry the function, a measure of head diversity must be anchored to an identifiable object. Heads can be characterized by a representative direction on a hypersphere, and the intuition of separating them is formalized with *spherical codes* (section 3); but the correct carrier is not the geometry of weights, but the *patch-response signature* that the head computes, the first singular vector of $\boldsymbol{C}_h^P = \boldsymbol{A}_h \boldsymbol{V}_h \boldsymbol{W}_O^{(h)}$, gauge-invariant. That signature is the instrument with which this work measures functional diversity, and it turns out to be head-specific.

Contribution. This work is an analysis, not a proposed method. The geometric apparatus is reduced to what the thesis uses —the notion of angular separation and the closed-form simplex (section 3)—; the appendix measures the gauge orbit on the real weights (appendix A).

1. A **dissociation across three intensities**: under the free gauge the geometry drifts up to 0.91 while the function remains intact to machine precision, invariance verified across the three columns; under the soft probe the geometry moves 23 to 33 degrees toward the simplex without the functional similarity separating from the between-seed noise, in the two finetuned columns, with the sign of the shift inverted between them; and under the hard variant, which imposes the simplex by construction, the geometry is pinned and the function does suffer —s_{func} shifts +0.031 and +0.052, same sign in both architectures, at a cost of 4.4 and 5.6 accuracy points—. The geometry-function coupling is one-directional (section 6.2).

2. A **closed-form impossibility**, with its decision consequence: under the gauge of the value-output sector, the orbit of $\boldsymbol{v}_1(\boldsymbol{W}_O^{(h)})$ is the full unit sphere of its row space, and the angle between two heads spans the whole interval allowed with the function fixed, so that no similarity criterion on $\boldsymbol{v}_1(\boldsymbol{W}_O)$ —redundancy-based pruning, dominant-direction interpretation, routing— admits a positive certified radius; the OV circuit is what is identifiable. The consequence is measurable: the pruning decision that geometry produces changes under gauge in more than 90 % of cases, both on a vision-finetuned ViT and, by the same closed form and without any training, on a pretrained language transformer (section 4 and table 3).

3. The **relocation toward $Q \cdot K$, bounded to regime**: the organization of attention that distinguishes one head from another shifts with depth toward the query-key comparison, where $\boldsymbol{v}_1(\boldsymbol{W}_O)$ does not participate; the shift is clean in the anchor, sustained but qualified in a second scale, and absent in a frozen self-supervised backbone, so that it is supervised finetuning, not global attention by itself, that produces it (section 6.4).

4. A **gauge-invariant instrument**, the positive side of the same coin: the response signature $\boldsymbol{v}_1(\boldsymbol{C}_h^P)$ lives in the same $\mathbb{R}^d$, is gauge-invariant and head-specific, and its decision survives the reparametrization that breaks that of the weight proxy —a certified impossibility and a certified possibility, distinguished by a single criterion (sections 4.2 and 6.5 and table 3)—.

That it is the correct diversity measure does not imply that it is the best criterion for *selecting* what to prune: at this budget, no criterion —gauge or functional— beats the random floor (table 7).

Data scope. Results are established on a ViT-B/16 finetuned on ImageNet-100 with five seeds —the anchor column— and contrasted on two more columns with three seeds each: a finetuned ViT-L/16 and a frozen DINOv2 with a linear head (table 1). The contrast did not turn out uniform: gauge and residual reproduce across the three columns, inertness across the two finetuned ones, and relocation only in them —clean in the anchor, qualified in ViT-L, absent in the frozen one—. Not every support applies to every column: inertness requires training with the probe and is measured on ViT-B and ViT-L; on the frozen column only the supports that need no retraining run (gauge, relocation, residual). Non-identifiability is closed-form and does not depend on architecture; reading relocation as a depth-dependent shift is correlational and indicative, not causal. Generalization to other head ratios and dimensions is discussed as a limit (section 7), not as a result.

2 Related work

The geometric proxy of the output projection. Reading the role or the redundancy of a head from the dominant direction of its output projection is the cheapest, most static proxy imaginable: it needs no forwards or gradients, it operates directly on the checkpoint. This work closes that route ahead of time more than it corrects a practice already widespread in the reviewed literature. State-of-the-art pruning methods in this setting do not use it: HEART-ViT [15] prunes heads and tokens by Hessian sensitivity on ViT-B/16 on ImageNet-100, Yang et al. [14] prune with global Hessian saliency, and the head-redundancy analysis with the longest track record reads gradients or activations, not weight geometry [8, 9]. The most recent mechanistic interpretability, when it does read weight geometry per head, converges instead toward the gauge-invariant object rather than the isolated dominant direction: Ahmad et al. [17] decompose by SVD the composite OV circuit $\boldsymbol{W}_v \boldsymbol{W}_O$ (and the QK circuit), not $\boldsymbol{W}_O$ on its own; Franco et al. [18] likewise decompose the circuit $\boldsymbol{W}_q^\top \boldsymbol{W}_k$; and Yamagiwa et al. [19] compare *full-rank* subspaces of the output projection —precisely the object that is here certified invariant under the gauge of the value-output sector (proposition 1)—, not its isolated dominant direction. The correction this work contributes is, accordingly, a corollary of the measured geometry-function dissociation (section 6.2), not the thesis: it protects the cheapest shortcut —reading $\boldsymbol{v}_1(\boldsymbol{W}_O)$ alone— that the more careful practice is already moving away from, without claiming to have corrected anyone in particular.

The gauge orbit as a resource, not only a warning. The gauge freedom that is here certified as a source of non-identifiability (proposition 1) is the same one that rotational quantization actively exploits as a design resource. QuaRot [20] and SpinQuant [21] insert orthogonal rotations that leave the model's output intact —their term is "computational invariance"— to remove outliers before quantizing; SpinQuant's rotation R_2 is applied exactly to the value-output pair ($\boldsymbol{W}_v^{(h)} \to \boldsymbol{W}_v^{(h)} R_2$, $\boldsymbol{W}_O^{(h)} \to R_2^{-1} \boldsymbol{W}_O^{(h)}$ for every head h of the layer, with R_2 orthogonal), the same gauge of proposition 1 restricted to the orthogonal subgroup $O(d_h) \subset GL(d_h)$ and, unlike the orbit certified here, shared across the heads of the layer rather than independent per head. That this restriction of the orbit has measurable operational consequences in the compression literature is independent evidence that the choice of gauge is not only noise to warn against, but a design space. Quantizing the OV circuit along that orbit, and measuring how its error moves with the chosen point, falls outside this work but is the most direct design continuation; together with component attribution on $Q \cdot K$, it sets the course for future work (section 7).

Head diversity by regularization: the direct baseline. Head diversity has been promoted with penalties of several kinds. The closest antecedent intervenes on a signal tied to behavior: an orthogonality regularizer on the input gradients of each head increases the diversity of the learned features and improves out-of-distribution generalization [10]. That approach alters the function —and gains OOD from it—, against which this work claims a simpler intervention: angularly separating the *patch-response signature*, with no input gradients or second order, using the same simplex machinery as the discarded weight regularizer. The contrast with intervening on $\boldsymbol{W}_O$ —which does not alter the function— is consistent with the organization that distinguishes one head from another residing in the computed response, not in the dominant geometry of the weights.

Depth-wise structure in attention. Functional stratification by depth in Transformers is documented. Pan et al. [13] decompose the $Q{\cdot}K$ interaction by SVD of $\boldsymbol{W}_q^{\top}\boldsymbol{W}_k$ and document a transition from perceptual grouping (early layers) to contextualization (deep ones). Yang et al. [14], when pruning, observe that the output projection and $Q{\cdot}K$ have distinct importances and a non-monotonic trend in attention diversity by layer. This depth-wise structure is not a contribution of this work, but context: it frames the seam at layer 5, where the organization of attention leaves residual writing and moves to the query-key comparison, and the inversion toward $Q{\cdot}K$ that we document (section 6) is a neighbor of those observations.

Redundancy and roles of multi-head attention. The analysis of multi-head attention has shown that many heads are dispensable [8] and that they fulfill specialized, identifiable functions [9]; a head can be important for the task without the geometry of its output projection tracing that role. Whether that specialization is an emergent property of training or a static property recoverable from the weights is the question that this work's results address (sections 6 and 7).

Neural collapse and fixed classifiers. The *neural collapse* phenomenon [2] describes how, in the terminal phase of training, class means and classifier vectors converge to a tight equiangular frame; later work [3] analyzes it under the unconstrained-features model. That regularity has motivated *fixed* classifiers with predetermined geometry [4]. The present work inherits the idea —handing over the geometry instead of waiting for it— but applies it to the heads' patch response, not to their weights.

Angular margins. SphereFace [5], CosFace [6], and ArcFace [7] introduced margins on the angle between the feature and the class prototype. They operate on an angular geometry, but estimate the prototypes during training; here the angular target is fixed with guaranteed separation and applied to each head's response signature.

3 Theoretical framework

This section keeps only what is essential to situate the intervention: the notion of angular separation on the hypersphere and the regime in which the model operates, which fixes the regularizer's target $\theta^{\star}$. In the loose regime in which every column of this work lives, the optimal code is closed-form and requires no generation machinery.

3.1 The hypersphere: a prior intuition

When several identical, mutually repelling elements are confined to a closed surface, they do not distribute at random: they settle into the rest state where the net force vanishes. Electrons on a conducting sphere —the problem Thomson posed in 1904— and the seeds in a sunflower's head obey the same principle: each element occupies the position that maximizes its angular distance to the rest. That rest state is not an act of computation, but the residue of a repulsion that

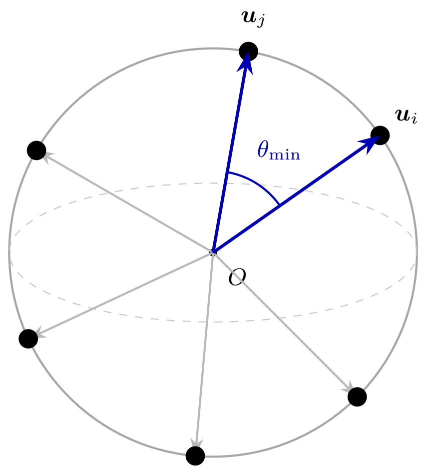

Figure 1: Two-dimensional schematic of a spherical code on $\mathbb{S}^{d-1}$. Each unit direction $\boldsymbol{u}_i$ corresponds, in this work, to the representative direction of an attention head —$\boldsymbol{v}_1(\boldsymbol{W}_O)$, the object the probe separates— or to its gauge-invariant response signature, the measurement instrument. Code quality is measured by the minimum angular separation $\theta_{\min}$, the smallest of the pairwise angles; a good code maximizes it.

has been left to act to the end. From an entirely continuous process, however, integer quantities emerge —the terms of the sunflower spiral's Fibonacci sequence—; the number of elements is not imposed from outside, but is inscribed in the geometry of the arrangement. A neural network, when separating its classes in representation space, reproduces that same rest state: the directions associated with each class move apart until they form an arrangement of maximum angular separation, a phenomenon that the recent literature labels *neural collapse*. The difference from physical systems opens an opportunity: where nature requires the entire relaxation process to reach that configuration, a model can be handed it fixed in advance. The following pages formalize that configuration on the unit hypersphere $\mathbb{S}^{d-1}$ and make precise the notion of minimum angular separation between pairs of directions.

3.2 Spherical codes and angular separation

Definition 1 (Spherical code)**.** *A spherical code of size K in dimension d is a set $\mathcal{C} = \{\boldsymbol{u}_1, \dots, \boldsymbol{u}_K\} \subset \mathbb{S}^{d-1}$ of unit directions. Its minimum angular separation is*

$$\theta_{\min}(\mathcal{C}) = \min_{i \neq j} \; \arccos(\langle \boldsymbol{u}_i, \boldsymbol{u}_j \rangle). \tag{1}$$

The goal is to find $\mathcal{C}^\star$ that maximizes $\theta_{\min}$. In the loose regime ($K \leq d+1$), the only one this work visits (K=12 or 16 against d=768 or 1024), the signed optimum is the regular simplex and is closed-form: its Gram matrix factorizes —ones on the diagonal, $-1/(K-1)$ off it— and embeds in $\mathbb{R}^d$ by rotation. The hard variant's target code (section 4.3) is that configuration; its construction is deterministic and reproducible without any descent.

3.3 Two regimes and the regularizer's target

The angular separation attainable by a code of K directions in $\mathbb{S}^{d-1}$ depends on the relation between K and d, and on whether the directions carry a sign. For signed directions, summing the K unit directions and observing that the squared norm is non-negative,

$$0 \leq \left\| \textstyle\sum_h \boldsymbol{u}_h \right\|^2 = K + \textstyle\sum_{h \neq h'} \langle \boldsymbol{u}_h, \boldsymbol{u}_{h'} \rangle, \tag{2}$$

the mean pairwise cosine is at least $-1/(K-1)$, and the maximum cosine cannot fall below its mean: $\theta_{\min} \leq \arccos(-1/(K-1))$ for *any* configuration, not only the equiangular ones. In the *loose regime* ($K \leq d+1$) the bound is attained —with equality exactly when the sum vanishes and the configuration is equiangular— by the regular simplex, which is therefore optimal [1]. For directions defined *up to sign* —singular vectors, elements of projective space— the relevant similarity is $|\cos|$ and the unsigned angle lives in $[0°, 90°]$: if $K \leq d$, the maximum is attained by the orthogonal frame (90°), and the simplex read without sign ($\arccos(1/(K-1))$) falls below it. The role of each value in the regularizer —signed optimum, unsigned threshold— is made precise in section 4.3. The saturated regime ($K > d+1$) is not visited in this work.

3.4 Connection to neural collapse

Neural collapse [2] establishes that, at the end of training, the classifier vectors converge to a tight equiangular frame: a configuration of maximum angular separation for $K \leq d+1$ classes. The network, therefore, spontaneously tends toward the spherical code. This work's intervention does not wait for that geometry to emerge, but regularizes it from the start on the patch response. In the loose regime in which the model operates, neural collapse's equiangular frame and the regularizer's target simplex are the same configuration.

4 The value-output gauge and the identifiable signature

The practice this work examines reads a direction from the weights: the first singular vector $\boldsymbol{v}_1(\boldsymbol{W}_O^{(h)})$ of each head's output projection, taken as its representative direction for redundancy-based pruning, dominant-direction interpretation, or routing. This section fixes what is identifiable in that sector and what is not, and derives the signature on which the work's measures rest.

4.1 The OV circuit and the non-identifiability of $\boldsymbol{v}_1(\boldsymbol{W}_O)$

A head's value-output sector factorizes its contribution to the residual stream into two matrices, $\boldsymbol{W}_v^{(h)} \in \mathbb{R}^{d\times d_h}$ and $\boldsymbol{W}_O^{(h)} \in \mathbb{R}^{d_h \times d}$, but that factorization is not unique. For any invertible $\boldsymbol{R} \in GL(d_h)$, the substitution

$$\boldsymbol{W}_v^{(h)} \to \boldsymbol{W}_v^{(h)}\boldsymbol{R}, \qquad \boldsymbol{W}_O^{(h)} \to \boldsymbol{R}^{-1}\boldsymbol{W}_O^{(h)}, \tag{3}$$

leaves the function intact, because the product $\boldsymbol{W}_v^{(h)}\boldsymbol{W}_O^{(h)}$ —the *OV circuit*— does not change. The OV circuit is the sector's only identifiable object: any property of the head that governs its function must be a function of it. The dominant direction $\boldsymbol{v}_1(\boldsymbol{W}_O^{(h)})$ is not. Under eq. (3) it shifts —the orthogonal subgroup of $\boldsymbol{R}$ rotates it, and the non-orthogonal part, present in every generic gauge, pulls it off its line— so that two functionally identical models can exhibit any angular separation between their $\boldsymbol{v}_1(\boldsymbol{W}_O)$. The claim admits a closed form and an exact bound:

Proposition 1 (Null certified radius of the $\boldsymbol{v}_1$ readout)**.** *Let $\boldsymbol{W}_O^{(h)} \in \mathbb{R}^{d_h\times d}$ be full rank and $\mathcal{S}_h$ its row space. For every unit direction $\boldsymbol{w} \in \mathcal{S}_h$ there exists $\boldsymbol{R} \in GL(d_h)$ with $\boldsymbol{v}_1(\boldsymbol{R}^{-1}\boldsymbol{W}_O^{(h)}) = \pm\boldsymbol{w}$, and the substitution (3) leaves the OV circuit intact. Consequently, the angle between the dominant directions of two heads sweeps, under gauge, every angle allowed by the row spaces —from the first principal angle $\theta_1(\mathcal{S}_h, \mathcal{S}_{h'})$ up to 90°— with the function fixed, and no similarity threshold on $\boldsymbol{v}_1(\boldsymbol{W}_O)$ admits a positive certified radius.*

Proof. With $\boldsymbol{W}_O = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top$ and $\boldsymbol{M} = \boldsymbol{R}^{-1}\boldsymbol{U}\boldsymbol{\Sigma}$ we have $\boldsymbol{R}^{-1}\boldsymbol{W}_O = \boldsymbol{M}\boldsymbol{V}^\top$, whose right singular vectors are $\boldsymbol{V}\boldsymbol{Q}$ with $\boldsymbol{Q}$ the right singular vectors of $\boldsymbol{M}$. As $\boldsymbol{R}$ ranges over $GL(d_h)$, $\boldsymbol{M}$ ranges over all of $GL(d_h)$ and $\boldsymbol{Q}$ over every orthonormal basis of $\mathbb{R}^{d_h}$; the first column can be placed at any desired coordinate. The orbit of $\boldsymbol{v}_1$ is the full unit sphere of $\mathcal{S}_h$. □

The nuance that makes this non-trivial: the row space *is* gauge invariant; what moves is the direction within the subspace, not the subspace. The negative result is therefore fine-grained

—the subspace does carry information (section 6.4); the dominant direction within it does not—. Reading $\boldsymbol{v}_1$ is reading a choice of gauge, not a property of the function. section 6.1 measures this on the real weights: the separation between the $\boldsymbol{v}_1(\boldsymbol{W}_O)$ shifts with $\boldsymbol{R}$ while the output and the OV circuit remain fixed, and the attainable orbit is certified pair by pair (appendix A).

4.2 The gauge-invariant response signature

If the dominant direction does not carry the function, a measure of head diversity must be anchored to an identifiable object. The head's per-patch contribution to the residual stream,

$$\boldsymbol{C}_h^P = \boldsymbol{A}_h \boldsymbol{V}_h \boldsymbol{W}_O^{(h)} \in \mathbb{R}^{P \times d}, \tag{4}$$

—the product of the attention map $\boldsymbol{A}_h$, the value projection $\boldsymbol{V}_h$, and the output projection $\boldsymbol{W}_O^{(h)}$— is the response the head *computes* and writes into the residual stream, and it is gauge-invariant: $\boldsymbol{A}_h \boldsymbol{V}_h \boldsymbol{R} \boldsymbol{R}^{-1} \boldsymbol{W}_O^{(h)} = \boldsymbol{C}_h^P$. Its first right singular vector,

$$\boldsymbol{r}_h = \boldsymbol{v}_1(\boldsymbol{C}_h^P) \in \mathbb{S}^{d-1}, \tag{5}$$

—the *response signature*, the residual-stream direction in which the head deposits the most energy across patches— is well-defined across heads and lives in the shared residual stream $\mathbb{R}^{768}$, the same space as $\boldsymbol{v}_1(\boldsymbol{W}_O)$, which makes the weight readout and the function readout comparable.[1] This signature is the work's measurement instrument: it quantifies heads' functional diversity without the gauge ambiguity that invalidates the $\boldsymbol{v}_1(\boldsymbol{W}_O)$ readout. section 6.5 verifies that it is head-specific.

4.3 The regularizer on $\boldsymbol{W}_O$ as an inertness probe

To test whether intervening on the weight geometry moves the function, this work uses as a *probe* —not as a proposed method— the angular regularizer that the practice would use: separating the representative directions $\{\boldsymbol{v}_1(\boldsymbol{W}_O^{(h)})\}_{h=1}^{H}$ toward the simplex threshold. The set is a spherical code of H directions defined up to sign, and the term pushes their absolute cosines below the threshold,

$$\mathcal{R}_{\text{ov}} = \sum_{h \neq h'} \max\left(0,\ |\langle \boldsymbol{v}_1(\boldsymbol{W}_O^{(h)}), \boldsymbol{v}_1(\boldsymbol{W}_O^{(h')}) \rangle| - \cos\theta^\star\right), \tag{6}$$

with $\theta^\star = \arccos(1/(H-1)) \approx 84.78°$ the threshold inherited from the H-vector simplex (section 3.3). The absolute value is necessary: two heads with nearly antiparallel directions are equivalent up to sign, and the signed product would read them as well separated. The penalty vanishes across the whole band $|\cos| \leq 1/(H-1)$, so it demands sufficient separation, not a specific configuration. The probe is added to the cross-entropy with weight λ,

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda\, \mathcal{R}_{\text{ov}}, \tag{7}$$

and the empirical question is whether, upon separating those directions, the functional similarity between heads moves. section 6.2 answers that it does not, at the fixed intensity ($\lambda = 10^{-3}$).

5 Experimental setup

Dataset. We use ImageNet-100, the 100-class subset of ImageNet. A hundred classes provide enough angular richness for between-head diversity to be measurable, and the phenomenon under study depends on the relation between the number of heads and the dimension, not on the number of classes; ImageNet-100 is thus representative without the full scale.

[1] The singular vector is defined up to sign; comparisons between singular signatures therefore use the absolute value of the dot product.

Architectures and environment. The apparatus —the $GL(d_h)$ gauge of the value-output sector, the OV circuit, the $Q{\cdot}K$ interaction, and the $\boldsymbol{C}_h^P$ signature— is defined the same way across any global attention, so it is contrasted across three columns that vary the two relevant axes without reinterpreting a single metric (table 1). ViT-B/16 and ViT-L/16 change the head/dimension ratio (H from 12 to 16, d from 768 to 1024, with $d_h = 64$ constant); DINOv2 at ViT-L scale changes the training regime —self-supervised versus supervised— at that same scale. All three start from pretrained weights (`timm`); ViT-B and ViT-L are finetuned on ImageNet-100, and DINOv2 is analyzed frozen, with no finetuning at all, so its attention maps are those of pure self-supervised pretraining. The anchor architecture (ViT-B) is trained with **five seeds** ($\{42, \ldots, 46\}$) and ViT-L with three; the frozen column is a single realization (n=1 de facto: with no finetuning there are no seeds to move the weights or the maps) and its cells carry no band. n is reported per row. Between-seed variability is the noise floor against which effects are contrasted. The environment is a single 32 GB NVIDIA RTX 5090 GPU. Windowed attention —Swin and, in general, the non-global kind— is left out: non-identifiability would transfer (the value-output gauge is structural to any value-output factorization), but inertness and relocation are measured on objects that its hierarchical, local topology redefines —the map stops being global, depth stops being uniform— and they enter as an extension, not as a column of these tables.

Table 1: Contrasted architectures and outcome by support. ViT-B→ViT-L varies the H/d ratio at constant d_h (the "other ratio"); ViT-L→DINOv2 varies the regime at constant scale. n: seeds; the frozen column is a single realization, with no between-seed spread. Gauge: output invariance (fp64) with drift of $\boldsymbol{v}_1(\boldsymbol{W}_O)$; residual: layers with paired < null; crossover: layer of the sustained crossover $r_{Q\cdot K} > r_{\boldsymbol{W}_O}$ against the 24-layer thresholds fixed before the data; inertness (H1): only finetuned columns. ViT-L's crossover passes two of the three thresholds (jump +0.403, margin +0.366) and fails deep dominance (+0.375 against the $\geq$ 0.40 threshold) at its final layer; the conjunctive outcome is ambiguous and is reported as such (section 6.4).

Architecture	H/d	n	Regime	Gauge	Residual	Crossover	Inertness
ViT-B/16 (anchor)	12/768	5	sup., finetuned	yes	12/12	layer 5 (5/5)	bound
ViT-L/16	16/1024	3	sup., finetuned	yes	24/24	layers 8–11 (3/3)	bound
DINOv2 (ViT-L)	16/1024	1	SSL, frozen	yes	24/24	**does not transfer**	n/a

Metrics. Unsigned minimum angular separation between the representative directions $\boldsymbol{v}_1(\boldsymbol{W}_O)$, $\theta_{\min}$, measured per layer —the minimum over pairs within that layer— and reported as the mean across layers; angular redundancy between heads —mean $|\cos|$ over pairs of those directions—; functional similarity s_{func} —cosine between average attention maps—; top-1 accuracy. All are reported with mean and between-seed band, and each run's final values are extracted as the average of the last five converged epochs —the same window for every variant and metric— so that every cell is traceable to its `run.csv`.

Frozen probing set. The functional measures share a single validation subset, sampled once with a fixed seed (10 images per class, 1000 in total, stratified), so that sampling noise does not contaminate the per-layer comparisons.

5.1 Hypotheses and tests, fixed before the data

This work makes one claim —the dominant direction of the output projection does not carry the function— and spreads the confirmatory discipline across its measured supports. The non-identifiability piece (section 4.1) is closed-form and does not enter this discipline; the three empirical contrasts are pre-specified here, with their metrics and directions fixed.

H1 (inertness). The probe on $\boldsymbol{v}_1(\boldsymbol{W}_O)$ (eq. (6)) raises the angular separation up to the simplex threshold, $\theta_{\min} \to \theta^\star$, but the aggregate functional similarity does not drop below the base level:

$$\Delta s_{\text{func}} = s_{\text{func}}^{\lambda} - s_{\text{func}}^{\text{base}} \;\approx\; 0, \tag{8}$$

at the fixed intensity ($\lambda = 10^{-3}$). The prediction is the *absence* of an effect, not its presence, and it is judged against the between-seed noise floor: inertness holds if $|\Delta s_{\text{func}}|$ does not exceed the base band while $\theta_{\min}$ saturates. H1 requires training with the probe and is tested on the two finetuned columns (ViT-B and ViT-L); the frozen column does not participate by construction.

H2 (operational non-identifiability). Under the gauge-flip of the value-output sector (eq. (3)) applied without retraining, the drift of $\boldsymbol{v}_1(\boldsymbol{W}_O)$ grows with the gauge magnitude $\|\boldsymbol{R}\|$ until it saturates, while the output and the OV-circuit signature remain at the numerical floor. The test is the monotonicity of the weight drift versus its constancy in the circuit, over the strength grid.

H3 (relocation). The per-layer correlation between s_{func} and the $Q{\cdot}K$ interaction overtakes that of the $\boldsymbol{W}_O$ subspace at an interior layer, with $Q{\cdot}K$ going from near zero to clearly positive at that layer —the abrupt jump, not a crossing by margin— and growing monotonically toward the deep layers. The crossover is reproducible *across architectures*, by seed within each one and across the three scales and regimes, and is indifferent to the probe. In the anchor the crossover falls at layer 5; for the new columns only its *existence and uniqueness* are pre-specified —an abrupt jump at an interior layer, with monotonic growth toward the deep layers— not its index: a 24-layer column need not cross "at layer 5". The response signature is, moreover, verified to be head-specific against a **permutation null** (section 6.5).

All three are read with their own band and with each architecture's own n. The confirmatory discipline rests on the *reproducibility of the sign and of the crossover*, not on inferential power tests: H1 —absence of movement in s_{func}— is required to be reproducible in the two finetuned columns, and H3 —crossover toward $Q{\cdot}K$— in all three, so that "five out of five in a single one" is elevated to its reproduction across two scales and two regimes, a firmer criterion than the power that two fewer seeds sacrifice in the new architectures. The prior exploratory smoke run fixed the noise band; the values are confirmed here on clean ImageNet-100 with the frozen set.

Outcome of the confirmatory discipline. H2 and the permutation null reproduce across the three columns; H1 in the two finetuned ones, with the sign of the shift inverted between them. H3 does not reproduce under the pre-specified terms: it holds in the anchor, remains ambiguous in ViT-L by one of the three thresholds, and does not hold in the frozen column. The criterion is applied exactly as it was fixed and the outcome is reported without readjusting it (section 6.4).

6 Results: the dominant direction does not carry the function

Results close over five seeds on a ViT-B/16 finetuned on ImageNet-100, and support a single claim with four supports: the dominant direction of the output projection does not carry the function of the heads. Its starting piece is closed-form and requires no retraining: the direction $\boldsymbol{v}_1(\boldsymbol{W}_O)$ —the one redundancy-based pruning and dominant-direction interpretation read— is not identifiable, because a gauge transformation of the value-output sector moves it at will without touching the function (section 6.1). On that basis, the angular probe operating on that geometry attains the separation with no accuracy cost but does not move the functional similarity (section 6.2), and the organization of attention that distinguishes one head from another relocates with depth toward the $Q{\cdot}K$ interaction, where that direction does not participate (section 6.4). A final diagnostic, also without retraining, verifies that the gauge-invariant response signature (section 4.2) is head-specific (section 6.5).

6.1 The value-output gauge: $\boldsymbol{v}_1(\boldsymbol{W}_O)$ is not identifiable

A head's contribution to the residual stream is invariant under a gauge freedom of the value space: for any $\boldsymbol{R} \in GL(d_h)$, the substitution $\boldsymbol{W}_v \to \boldsymbol{W}_v\boldsymbol{R}$, $\boldsymbol{b}_v \to \boldsymbol{b}_v\boldsymbol{R}$, $\boldsymbol{W}_O \to \boldsymbol{R}^{-1}\boldsymbol{W}_O$ leaves the function intact, because $\boldsymbol{W}_v\boldsymbol{W}_O$ —the OV circuit, the sector's only identifiable object— does not change. The dominant direction $\boldsymbol{v}_1(\boldsymbol{W}_O)$ —its first right singular vector, the writing direction into the residual stream—, in contrast, shifts with $\boldsymbol{R}$: the gauge's orthogonal subgroup preserves it, and it is the non-orthogonal part of $\boldsymbol{R}$, present in every generic transformation, that moves it. Applying that gauge to the finetuned base model, without retraining, we measure what each method reads (forward passes and signatures in fp64): the output does not change —relative error $\leq 2 \times 10^{-13}$ across the 420 layer×seed×strength combinations—, but the separation between the $\boldsymbol{v}_1(\boldsymbol{W}_O)$ —the quantity redundancy-based pruning compares— shifts, while the OV-circuit signature —its first singular vector, by exact SVD— stays twelve orders of magnitude below ($\leq 4.2 \times 10^{-13}$), at the double-precision floor.

The shift in $\boldsymbol{v}_1(\boldsymbol{W}_O)$ is not the accident of one specific transformation: it grows with the gauge magnitude $\|\boldsymbol{R}\|$ and saturates toward orthogonalization —from 0.05 with an $\boldsymbol{R}$ barely different from a scalar multiple of the identity, it climbs through 0.35 and 0.66, and settles at $\sim$0.83 (the ceiling of $1 - |\cos|$) from $\|\boldsymbol{R}\| \approx 1$ onward—, while the OV circuit's stays at $\sim 3 \times 10^{-13}$ whatever the strength (table 2). It is not "one gauge moves it", but "every generic gauge moves it, within the $|\cos|$ ceiling, and none moves the circuit". This is the closed-form demonstration that reading or reorganizing $\boldsymbol{v}_1(\boldsymbol{W}_O)$ —redundancy-based pruning, dominant-direction interpretation, routing by representative vector— operates on gauge noise. And it makes the inertness of the regularizer in section 6.2 —which optimizes that same non-identifiable object— plausible, without proving it: that an intervention on it does not move functional similarity through any indirect channel is an empirical fact, measured separately, not a consequence of the gauge.

Table 2: Premise, value-output gauge-flip on the base model, without retraining and in fp64 (mean ± standard deviation over the 60 combinations of 5 seeds × 12 layers per strength). Drift is the mean $1 - |\cos|$ between the signature before and after the gauge; $\|\boldsymbol{R}\|$ measures the deviation of $\boldsymbol{R}$ from a scalar multiple of the identity. The drift of $\boldsymbol{v}_1(\boldsymbol{W}_O)$ grows with gauge strength; that of the OV circuit does not: it stays at the double-precision floor, twelve orders of magnitude below, constant and independent of strength.

$\|\boldsymbol{R}\|$	Output error	Drift $\boldsymbol{v}_1(\boldsymbol{W}_O)$	OV-circuit drift
0.06	1.1×10^{-15}	0.05 ± 0.04	2.9×10^{-13}
0.13	1.1×10^{-15}	0.15 ± 0.09	2.9×10^{-13}
0.25	1.1×10^{-15}	0.35 ± 0.16	2.9×10^{-13}
0.50	1.1×10^{-15}	0.66 ± 0.15	2.9×10^{-13}
1.00	8.0×10^{-15}	0.83 ± 0.04	2.9×10^{-13}
2.00	8.5×10^{-15}	0.83 ± 0.05	2.9×10^{-13}
4.05	2.9×10^{-14}	0.82 ± 0.07	2.9×10^{-13}

The drift is not abstract: the decision the practice would make changes under the gauge. Materializing the two typical decisions —the most redundant pair of heads and the top-3 pruning set by mean redundancy— on every layer and seed, and applying five gauges per layer at saturated strength ($\|\boldsymbol{R}\| \approx 1$), the decision by $\boldsymbol{v}_1(\boldsymbol{W}_O)$ changes pair in 92.7 % of cases and retains on average 0.38 of the top-3 (per layer, between 0.24 and 0.69); the same decision taken with the gauge-invariant response signature never changes pair and retains the full top-3 (table 3). Two functionally identical models, under the weight criterion, prune different sets of heads.

The instability is not a property of vision or of finetuning: it reproduces, by the same closed form and without training anything, on a pretrained language transformer. On Pythia-410M [16] (GPT-NeoX, 24 layers, 16 heads, pure multi-head attention) the same value-output gauge, at an equivalent pruning budget (k=4 of 16, the same relative 25 % as the anchor's top-3 of 12), changes the most redundant pair in 90.0 % of cases and retains only 0.229 of the pruning set (table 3). The response signature requires forward passes with a probe and is not ported to this column; the comparison is limited to the weight criterion.

Table 3: Decision under gauge, without retraining. ViT-B: 5 seeds × 12 layers × 5 gauges at $\|\boldsymbol{R}\| \approx 1$, top-3 of 12 (25 %). Pythia-410M: 24 layers × 5 gauges at the same saturated strength, top-4 of 16 (same relative 25 %; frozen column, n=1 de facto); the response signature is not measured on the language column (it requires forward passes with a probe, out of scope). Fraction of cases in which the most redundant pair changes and mean overlap of the pruning set with the no-gauge decision, by criterion. Thresholds fixed before the data: weights unstable if the pair changes in ≥ 40 % of cases or the mean overlap < 0.67; signature stable if the overlap ≥ 0.95 and the pair changes < 5 %.

Architecture	Criterion	Pair changes (%)	Top-k overlap
ViT-B/16 (anchor, k=3)	$\boldsymbol{v}_1(\boldsymbol{W}_O)$ (weights)	92.7	0.378
ViT-B/16 (anchor, k=3)	Signature $\boldsymbol{v}_1(\boldsymbol{C}_h^P)$	0.0	1.000
Pythia-410M (language, k=4)	$\boldsymbol{v}_1(\boldsymbol{W}_O)$ (weights)	90.0	0.229
Pythia-410M (language, k=4)	Signature $\boldsymbol{v}_1(\boldsymbol{C}_h^P)$	not measured	not measured

proposition 1 is further certified pair by pair on the real weights (appendix A): across the five seeds, the orbit's floor —the first principal angle between row spaces— drops as low as 14.7° depending on the pair, the simplex threshold $\theta^\star = 84.78°$ falls within the attainable band of the 3960 pairs measured (66 pairs × 12 layers × 5 seeds), and a descent restricted to the row spaces finds, across the 60 layer×seed combinations, a joint configuration of the twelve directions with $\theta_{\min} \geq \theta^\star$ (worst case 84.79°). *Total* compliance of the regularizer therefore lies within the gauge orbit: there is a path of exactly zero functional cost toward it. Every observed angle between dominant directions is a point on a band tens of degrees wide that the gauge sweeps with the function fixed; the specific value is a choice of gauge, not of function.

6.2 The regularizer on $\boldsymbol{W}_O$ separates at no cost, but does not move the function

The weight carrier is the representative direction $\boldsymbol{r}_h = \boldsymbol{v}_1(\boldsymbol{W}_O^{(h)})$, the output projection's first right singular vector, and the regularizer is that of eq. (6) on that carrier. Finetuning alone separates the heads only partially: $\theta_{\min}$ stalls at 51.96°, well below the simplex threshold ($\theta^\star = 84.78°$). The stall does not indicate a packing obstruction: with K=12 $\leq d$=768 the unsigned optimum is the orthogonal frame at 90°, with 756 dimensions of slack, and the admissible band $|\cos| \leq 1/11$ is enormous; it is a non-obstruction certificate, and the 51.96° turns from raw data into directed evidence. The soft regularizer drives that separation to 85.01° —it clears the threshold and keeps drifting toward the orthogonal frame— and reduces angular redundancy from 0.172 to 0.034, with a 0.10 pp accuracy drop that falls *within* the base's between-seed top-1 band (0.19 pp): that, not a nominal zero, is what "at no cost" means. The hard variant, which reprojects onto the simplex after every step, is pinned to it (84.77°) —below the soft one, which clears it without any code at all: the hinge drifts freely toward the orthogonal frame while the reprojection fixes it to a suboptimal configuration— and pays 4.40 pp —uniform across seeds (range 0.86 pp), 23 times the band— and destabilizes accuracy within each run (intra-run deviation 0.0022 versus 0.0013 for base); the soft variant not only matches the hard one in separation, it beats it in de-correlation. table 4 records the final state.

The pattern repeats in the deep column: the hard variant pays 5.6 pp —uniform across seeds— and shifts s_{func} +0.031, thirty-one times ViT-L's base band, while the soft variant reaches the same $\theta_{\min}$ at no measurable cost. The hard variant's shift keeps its sign and order of magnitude across architectures; the soft one's inverts them and stays in the noise.

On s_{func}, the heart of H1: the shift between variants is of the order of the between-seed spread ($\Delta = +0.0013$, 95 % CI $[-0.0011, +0.0037]$, which includes zero), with no detectable separation from zero across five replicates, with a possible sign bias smaller than that spread. The result is a *bound*, not a null. Its sensitivity analysis: the full between-seed band of base s_{func} is 0.0041, inflated by two atypical runs (one high, one low, of opposite signs); the core without them is 0.0017. The effect (0.0013) falls within both measures, and the seed-paired reading agrees with the reading against the base mean except for the two atypical seeds —the dual pairing that detects them—, so that effect, between-seed noise, and run noise are of the same size and none separates out. The hard variant, by contrast, does NOT leave s_{func} still: it shifts it +0.052 upward —12.8 times the band, uniform across the five seeds—, that is, attention maps become *more* similar across heads while the weight geometry ends up maximally separated. The asymmetry is itself a result: two interventions on the same geometry reach almost the same $\theta_{\min}$ —gradient descent on the penalty for one, forced reprojection with no backpropagation for the other— and one is inert while the other drags the function along, depending on the mechanism. The inertness H1 establishes is that of the *gradient probe* at the fixed intensity, not that of any manipulation of that geometry: forcing the weights off the optimizer's trajectory costs accuracy and moves functional similarity. Numerical detail of the soft probe: the SVD gradient degrades near convergence; $\theta_{\min}$ saturates around epoch 6 but s_{func} drifts up to ~ 25 and stabilizes afterward (epoch-to-epoch deviation 0.0007), which is why the metrics are extracted from the converged window (section 5).

Table 4: Premise, part I: ablation of the angular regularizer on the geometry of $\boldsymbol{W_O}$ (clean ImageNet-100, 30 epochs, mean ± standard deviation across 5 seeds; per-run values averaged over the converged window). $\theta_{\min}$ is the unsigned minimum angular separation between representative directions, per layer and reported as the mean across layers (threshold 84.78°); redundancy is angular (mean $|\cos|$ between directions). The soft variant separates at a cost within the between-seed band ($0.10 < 0.19$ pp) and a shift in s_{func} bounded by that same spread; the hard variant pins the code, paying 4.40 pp and moving s_{func} +0.052 upward: two paths to the same $\theta_{\min}$, one inert and one not, depending on the mechanism.

Configuration	$\theta_{\min} \uparrow$	Redundancy $\downarrow$	Top-1 $\uparrow$	s_{func}
Base ($\lambda_A = 0$)	$51.96° \pm 0.26$	0.1724 ± 0.0030	0.9284 ± 0.0019	0.6945 ± 0.0041
Soft ($\lambda_A = 10^{-3}$)	$85.01° \pm 0.08$	0.0338 ± 0.0009	0.9274 ± 0.0017	0.6958 ± 0.0028
Hard (reprojection)	$84.77° \pm 0.00$	0.0909 ± 0.0000	0.8844 ± 0.0021	0.7466 ± 0.0049

6.3 The fixed intensity and the dependence on λ

Inertness is established at the fixed intensity $\lambda = 10^{-3}$ (section 6.2): $\theta_{\min}$ saturates without s_{func} dropping below the base level. Characterizing the dependence on λ over a wider range remains as an extension, and would require the same reading apparatus —five seeds per intensity, against the between-seed noise floor— that the support at $\lambda = 10^{-3}$ already consumes. The relevant caveat: at much higher intensity the probe could stop being an inert perturbation and start deforming the representation, so indifference is not asserted beyond the measured intensity.

6.4 The substrate of attention relocates toward $Q{\cdot}K$ with depth

Beyond the non-identifiability of v_1 (section 6.1), a layer-wise characterization of the finetuned model shows where the organization of attention resides and why the geometry of $\boldsymbol{W_O}$ is inert.

For every layer, functional similarity s_{func} —cosine between average attention maps— is correlated with two geometric candidates: the column subspace of $\boldsymbol{W}_O$ (similarity between the 768×64 per-head spaces via principal angles) and the $Q \cdot K^\top$ interaction (cosine between the pre-softmax logits). table 5 reports both together with the sole dominant direction v_1; fig. 2 plots them per layer with their between-seed band.

The pattern is an inversion. In the early layers the subspace of $\boldsymbol{W}_O$ correlates with the function while $Q \cdot K$ does not; from layer 5 onward the order flips and $Q \cdot K$ takes over, growing up to ~ 0.9 in the deep layers. The crossover $r_{Q\cdot K} > r_{\boldsymbol{W}_O\text{-subsp.}}$ occurs at layer 5 in all five seeds, with $Q \cdot K$ switching on abruptly between layer 4 and 5 ($r_{Q\cdot K}$ from 0.026 to 0.51). The inversion is asymmetric —$Q \cdot K$ goes from ≈ 0 to ≈ 0.9 while $\boldsymbol{W}_O$ was barely above noise— and independent of the regularizer: under the soft variant the crossover stays at layer 5 and $r_{Q\cdot K}$ is indifferent to the regularization (maximum $|\Delta| = 0.043$). The dominant direction v_1 that the weight regularizer touches does not participate in the crossover: it is low or negative across almost the entire depth, tracing function only in the layer 3–4 transition. That trace is real in the gauge training fixed, but it is not identifiable (section 6.1): another gauge would erase it without touching the function, so it does not contradict the non-identifiability of v_1 —it illustrates it, it is precisely the point that reading via v_1 depends on the gauge—. This is the measured reason for the inertness: the weight regularizer optimizes an object that does not govern the organization of attention.

On ViT-L (n=3, 24 layers) the sustained crossover exists in all three seeds —layers 8, 11, and 8, at a relative depth of 33 to 46 %, comparable to the anchor's 42 %— and it clears two of the three thresholds: the jump at the crossover layer is +0.403 (threshold ≥ 0.30) and the margin +0.366 (threshold ≥ 0.15). Deep dominance lands at +0.375, below the ≥ 0.40 band, and the conjunctive outcome is accordingly ambiguous. The diagnosis reveals a localized cause: layer 23 drags down the minimum in all three seeds —+0.43, +0.41, and +0.28— while layers 16 to 22 clear it in every case (+0.61 to +0.63); in the anchor the last third showed no such dip. The final layer is not excluded from the threshold computation, fixed before the data; the threshold is reported as it came out, together with its cause.

Relocation does NOT transfer to the frozen column. On DINOv2 (24 layers, declared n=1), the thresholds fixed before the data fail three of four: the sustained crossover only appears from layer 22 onward, with a jump of +0.223 at that point (within the band), the margin is +0.009 (threshold ≥ 0.15, fails), and deep dominance does not exist —minimum $r_{Q\cdot K}$ of the last third -0.059 against the ≥ 0.4 threshold, fails—: in the pure self-supervised backbone, $r_{Q\cdot K}$ does not take off in any depth band. H3, as pre-registered for the three columns, does not hold in the frozen regime.

Relocation is thus bounded to the finetuned supervised regime: clean in the anchor, sustained but qualified in the deep column, absent in the frozen one. H3, pre-registered as reproducible across the three columns, does not hold under those terms.

6.5 The residual carrier tilts from inertness without decoupling

The gauge-invariant response signature (section 4.2) is defined on the contribution to the residual stream, not on $\boldsymbol{v}_1(\boldsymbol{W}_O)$; a prior diagnostic, without retraining, measures how far apart they are. Per layer, each head's computed signature $\boldsymbol{v}_1(\boldsymbol{A}_h \boldsymbol{V}_h \boldsymbol{W}_O)$ is compared with its static signature $\boldsymbol{v}_1(\boldsymbol{W}_O)$ —the *paired* angle—, against the permutation null —the computed signature of h versus the static one of $h' \neq h$—. Both live in the 64-dimensional row space of $\boldsymbol{W}_O$, so chance here is the permutation within that subspace, not that of $\mathbb{R}^{768}$; both signatures are computed by exact SVD.

The paired angle stays below the null at every layer and is never zero (table 6): the computed signature does genuinely tilt relative to the static one —value/attention content moves it, the residual carrier does not collapse onto the inert one, which was the go/no-go condition— but it remains more similar to its own head than to the rest, that is, head-specific. The gap is modulated by depth —from 24° at layer 1 to 39° at layer 2— with two irregularities reproducible across seeds: at layer 8 the null drops to 73.1° against the $\sim$86° of the rest —the static directions $\boldsymbol{v}_1(\boldsymbol{W}_O)$ of

Table 5: Premise, part II: Spearman correlation between s_{func} and two geometric candidates, per layer (base variant, mean across 5 seeds; $r_{Q\cdot K}$ does not depend on k). The subspace is measured at $k = 64$; v_1 is the sole dominant direction. The random baseline of the subspace at $k = 64$ is ≈ 0.25; that of $Q\cdot K$, ≈ 0. The crossover toward $Q\cdot K$ at layer 5 marks where the organization leaves residual writing and moves to the query-key comparison.

Layer	$r_{\boldsymbol{W}_O}$ (subsp.)	$r_{\boldsymbol{W}_O}$ (v_1)	$r_{Q\cdot K}$
0	+0.439	+0.185	−0.104
1	+0.456	+0.075	+0.189
2	+0.503	+0.094	+0.233
3	+0.470	+0.463	+0.212
4	+0.341	+0.223	+0.026
5	+0.150	−0.005	+0.510
6	+0.176	−0.096	+0.296
7	+0.069	+0.038	+0.720
8	+0.058	−0.168	+0.811
9	−0.198	−0.313	+0.907
10	−0.041	−0.340	+0.770
11	+0.089	−0.108	+0.832

different heads resemble each other there— and at layer 11 paired and null fall together (23.5° and 51.9°), with the whole pattern shifted downward. Both are properties of the static signature in the gauge that training fixed —not identifiable, like any $\boldsymbol{v}_1(\boldsymbol{W}_O)$ readout (section 6.1)—, so they are named as features of the diagnostic, not as redundancy or functional convergence. This is a diagnostic of the carrier, not the inversion of section 6.4, and it is reported as such.

Table 6: Computed/static diagnostic of the residual carrier, without retraining (mean $\pm$ standard deviation across 5 seeds, frozen set of 1000 images). Paired angle (computed signature of h vs. static of h) against the permutation null (vs. static of $h' \neq h$), in degrees. Paired below null at every layer, and never at zero, indicates that the residual carrier tilts from inertness without decoupling.

Layer	Paired	Permutation null
0	43.0 ± 1.2	77.2 ± 0.2
1	60.4 ± 1.2	84.4 ± 0.1
2	46.6 ± 2.5	85.6 ± 0.1
3	50.8 ± 3.1	87.0 ± 0.1
4	50.8 ± 4.2	87.5 ± 0.1
5	59.5 ± 3.3	86.7 ± 0.2
6	52.3 ± 2.7	85.6 ± 0.2
7	64.4 ± 3.7	85.3 ± 0.2
8	44.2 ± 1.2	73.1 ± 0.7
9	49.6 ± 3.7	86.6 ± 0.2
10	52.3 ± 2.7	82.8 ± 0.5
11	23.5 ± 2.2	51.9 ± 1.6

6.6 Similarity is not a pruning criterion; stability is what the instrument offers

A minimal benchmark bounds what the instrument buys and what it does not. With the same budget (k=2 heads per layer, 24 of 144) and the same pruning mechanism —zeroing out the head's columns in the output projection—, the most redundant set is pruned according to each criterion and the drop in top-1 is measured on the full validation set (table 7). No similarity

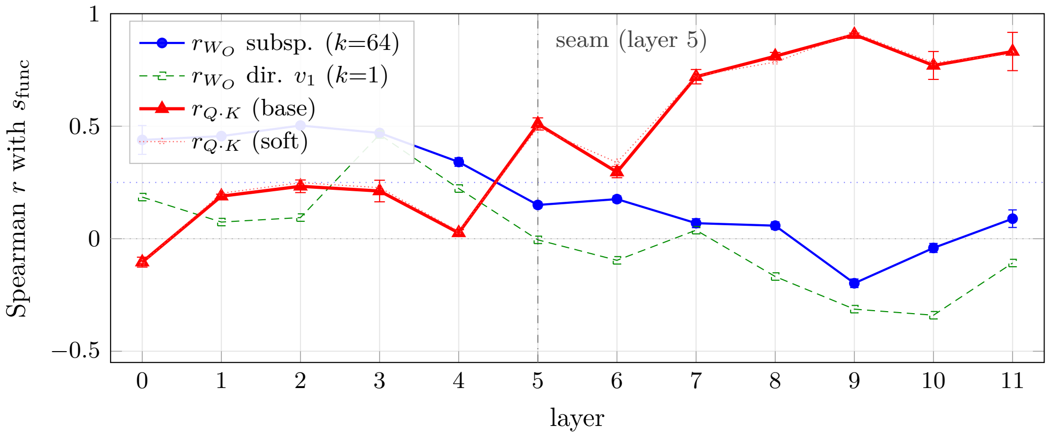

Figure 2: Premise, part II (visual of table 5): Spearman correlation between s_{func} and the geometric candidates per layer, mean across 5 seeds; bars are ±1 standard deviation. The subspace of $\boldsymbol{W}_O$ (k=64) correlates in the early layers and sinks with depth; $Q\cdot K$ starts in the noise, switches on abruptly at the layer 5 *seam* ($0.026 \to 0.51$), and dominates up to ~ 0.9. The soft variant's $Q\cdot K$ (dotted) falls on top of base: the crossover does not depend on the weight regularizer. The dotted lines mark the random baselines (≈ 0.25 for the subspace, ≈ 0 for $Q\cdot K$).

criterion beats the random floor: selection by weights falls within its band (3.13 ± 0.54 against 3.25 ± 0.19) —consistent with the fact that choosing by $\boldsymbol{v}_1(\boldsymbol{W}_O)$ is choosing by gauge noise—, and selection by signature does *more* damage than chance (4.26 ± 0.29). The diagnostic explains why: both criteria select heads with a large contribution to the residual (2.7× the mean energy of the remaining ones), and the ones the signature groups are also individually more costly (pruning half the pair at a time, the sum of the halves is already 3.72 against 2.90 pp for weights, with similar superadditivity in both, +0.8/+0.9 pp). Similarity —of gauge or functional— is not equivalent to dispensable redundancy: heads with similar signatures are central, not surplus. The applied statement, for practice, joins the two certificates: the decision by $\boldsymbol{v}_1(\boldsymbol{W}_O)$ admits no positive certified radius —the gauge fabricates it at will (proposition 1 and table 3)— and the decision by signature is invariant; but neither similarity selects what to prune better than chance at this budget. The instrument's advantage is the *stable decision* under reparametrization, not the selection; what to prune calls for importance criteria, outside the scope of this analysis.

Table 7: Pruning by criterion, without retraining (5 base seeds; k=2 heads per layer, 24 of 144; mean ± between-seed band; the random criterion averages three draws fixed per seed). The pruning mechanism is identical for the three criteria. No similarity beats the random floor; the instrument's role is decision stability, not selection.

Criterion	Pruned top-1	Drop (pp)
$\boldsymbol{v}_1(\boldsymbol{W}_O)$ (weights)	0.898	3.13 ± 0.54
Signature $\boldsymbol{v}_1(\boldsymbol{C}_h^P)$	0.887	4.26 ± 0.29
Random (floor)	0.896	3.25 ± 0.19

7 Discussion and limitations

The three arms, read together. Gauge, probe, and hard variant are not three loose results: they characterize, at three increasing intensities of intervention on the value-output sector, the coupling between that geometry and the function. The freest intervention —the gauge, which does not touch the function by construction— moves the geometry to the maximum at no cost; the intermediate intervention —the probe, which does train— also moves it at no cost; only the most aggressive one —the hard variant, which imposes the simplex at every step— pays in function. That the damage appears exactly where the intervention stopped being a gauge and started being an active constraint on the optimizer's trajectory is itself the datum distinguishing "reparametrization" from "manipulation", not a post-hoc reading. The asymmetry connects with the relocation: if the function does not live in $\boldsymbol{v}_1(\boldsymbol{W}_O)$, and the organization of attention does correlate with $Q{\cdot}K$ in the fine-tuned regime, the constructive question —which component governs the function, if not that direction— remains open, not answered; component-wise attribution over $Q{\cdot}K$ is the natural continuation (sections 6.2 and 6.4).

What the result claims and what it does not. The claim is negative and bounded: the geometry of the output projection —$\boldsymbol{v}_1(\boldsymbol{W}_O)$, the direction that redundancy-based pruning and interpretation by dominant direction read— does not carry the head's function. It is not identifiable under the value-output sector's gauge, separating it does not move functional similarity, and the organization of attention resides in another substrate. The work does not claim the opposite of that: it does not hold that an actionable weight carrier exists, nor that $Q{\cdot}K$ is the only substrate of the function —that reading is correlational—, nor that the response signature exhausts the available diversity. Intervening constructively on $Q{\cdot}K$, or on the response signature, is the natural continuation, not a result of this work.

Implications for pruning and interpretation. Part of head pruning decides redundancy by comparing representative directions of $\boldsymbol{W}_O$, and part of mechanistic interpretation reads a head's role from its dominant weight direction. Section 6.1 shows that those comparisons operate on a non-identifiable quantity: two functionally identical models exhibit different separations between their $\boldsymbol{v}_1(\boldsymbol{W}_O)$'s depending on the gauge, so that angular redundancy measured on weights is, in part, a reparametrization artifact, and the decision it produces does in fact change under gauge (table 3). The gauge-invariant response signature (section 4.2) offers the quantity those tasks need as a *measure*: head diversity anchored to the identifiable circuit, which does not move under reparametrization and is head-specific (section 6.5). With a measured caveat: stability is not a selection criterion —no similarity, of gauge or of function, beats the random floor as a pruning rule at a fixed budget (table 7); deciding *what* to prune still calls for importance criteria such as second-order ones [14, 15], on which this analysis takes no position—.

Emergent specialization, not structural. If heads specialize functionally [8, 9], the results suggest that specialization is *emergent* —a property of computed behavior— rather than *structural* —recoverable from the dominant geometry of the output weights—. A neighboring line of work that regularizes diversity over input gradients —a signal tied to behavior— alters the function and improves OOD generalization [10], whereas intervening on the weight geometry does not move it: the contrast places the effect in the functional signal, not in the static summary of the weights.

Limits. The non-identifiability is closed-form and does not depend on the architecture; inertness and relocation live within it, and are for that reason contrasted across three columns (table 1): two scales of the head/dimension ratio —ViT-B and ViT-L— and two training regimes —supervised and self-supervised, the latter with frozen DINOv2—. That the four supports also appear at another scale and under self-supervision separates "property of global attention" from "artifact of the supervised fine-tuning of a single column", a question a single architecture could not answer. The declared boundary is non-global attention: in Swin and window attention the value-output non-identifiability would indeed transfer —the value-output gauge is structural—, but the shifted-window local map, the relative position bias in the logits, and the stage-wise hierarchical depth require re-deriving s_{func} and the very notion of depth before measuring inertness or relocation; it remains an extension, not a hole in rigor. The reading of the relocation toward $Q{\cdot}K$ is correlational —Spearman between s_{func} and geometric candidates per layer—, not causal, and the choice of the first singular vector of $\boldsymbol{C}_h^P$ as the signature, against other summaries of the response, is a design decision whose contrast with patch-centering bounds its sensitivity without exhausting it. The deep-dominance threshold inherited the anchor's assumption that the last third of layers is homogeneous; ViT-L breaks it at its final layer, and the assumption is declared as a limit of the criterion, not corrected after the fact.

8 Conclusion

Three intensities of intervention on the value-output sector characterize the coupling between its geometry and the function of an attention head, and the coupling is one-directional. Under the free gauge, the dominant direction $\boldsymbol{v}_1(\boldsymbol{W}_O)$ —the one that redundancy-based pruning and interpretation by dominant direction read— shifts at will without touching the function; under the probe that separates it angularly up to the simplex, it does not either; only when that simplex is imposed by construction does the function suffer. The explanation for the inert extreme is closed-form: $\boldsymbol{v}_1(\boldsymbol{W}_O)$ is not identifiable under that gauge, which shifts it —up to 0.91 of drift— while the output and the OV circuit $\boldsymbol{W}_v\boldsymbol{W}_O$ remain at the numerical floor of double precision, an invariance verified identically across the three columns, including the frozen self-supervised one. The consequence is not caution: no similarity threshold on $\boldsymbol{v}_1(\boldsymbol{W}_O)$ admits a positive certified radius, and two functionally identical models prune different sets of heads —the most redundant pair changes in 92.7 % of cases on the fine-tuned ViT and in more than 90 % on a language transformer with no training at all, and the pruning-set overlap drops to 0.378 on the anchor—, while the same decision taken on the response signature remains intact.

Separating that direction angularly up to the simplex threshold does not move functional similarity. The probe raises $\theta_{\min}$ from 52° to 85° on the anchor and from 63° to 86° on the deep column, and the shift in s_{func} remains on the order of the between-seed dispersion in both, with the sign inverted between them —+0.0013 and −0.0019—. The same criterion distinguishes the opposite case: the hard variant, which imposes the simplex by construction, shifts s_{func} by +0.052 and +0.031 —twelve and thirty-one times the band, same sign in both— at a price of 4.4 and 5.6 points of accuracy. The soft probe's inertness is not, therefore, an incapacity of the measure.

The relocation of attention organization toward $Q\cdot K$ with depth is bounded to its regime. It is clean on the anchor —crossover at layer 5, five of five seeds, all three thresholds cleared—, sustained but qualified on ViT-L —crossover at layers 8 to 11 of 24, at a comparable relative depth, with a clean jump and margin and deep dominance failing at the final layer— and absent on the frozen DINOv2, where $r_{Q\cdot K}$ does not take off at any depth band. H3, as pre-registered for the three columns, does not hold: the reading the data admits is that supervised fine-tuning, and not global attention by itself, produces the relocation.

What remains, then, is a two-speed result, and it is worth reading it as such. Non-identifiability and inertness are closed-form, the former, and reproduced across two scales, the latter; the relocation is correlational and regime-dependent. For anyone pruning, interpreting, or routing by the geometry of $\boldsymbol{W}_O$, the implication does not depend on which of the two speeds one looks at: the quantity being read is the gauge choice that training fixed by accident, and the response signature $\boldsymbol{v}_1(\boldsymbol{C}_h^P)$ —invariant and head-specific across the three columns— is what takes its place.

Code and data availability

The code for the angular probe and all measurements, together with the gauge-orbit certification (appendix A), is available in the open repository `https://github.com/mmunozpl/angular-separation-vit`, archived on Zenodo at `https://doi.org/10.5281/zenodo.21630535`. The CSV files backing every table and figure of section 6 are published at `https://huggingface.co/datasets/ManuelPla/angular-separation-vit-results` (`https://doi.org/10.57967/hf/9743`), and the reproduction checkpoints —per seed and architecture— at `https://huggingface.`

co/ManuelPla/angular-separation-vit-checkpoints (https://doi.org/10.57967/hf/9742).

A The gauge orbit of $\boldsymbol{v}_1(\boldsymbol{W}_O)$, measured

This appendix numerically certifies proposition 1 on the actual weights of the five ViT-B base seeds, without retraining and on CPU.

Per-pair band. For each layer and each pair of heads (h, h'), the principal angles between the row spaces $\mathcal{S}_h$ and $\mathcal{S}_{h'}$ are computed (SVD of $\boldsymbol{B}_h^\top \boldsymbol{B}_{h'}$, with $\boldsymbol{B}$ the orthonormal row bases). The first principal angle is the floor of the joint orbit —no gauge can bring the dominant directions closer than that— and the ceiling is 90° (with $d_h + d_h \ll d$, orthogonal pairs always exist). Over 66 pairs × 12 layers × 5 seeds: global minimum floor 14.7°, and $\theta^\star = 84.78°$ within the band of the 3960 pairs.

Explicit realization. For one pair per seed, the gauge that carries each $\boldsymbol{v}_1$ to its target is constructed (the pair of directions with minimum angle between the two subspaces): the drift of the OV circuit stays at $\sim 2 \times 10^{-15}$ —invariance at machine precision— with the dominant directions shifted by tens of degrees.

Simultaneous certificate. The per-pair band is not enough for the regularizer, which imposes 66 coupled constraints on 12 directions. A descent restricted to the row spaces —free coordinates in the basis of each $\mathcal{S}_h$, penalty on $|\cos| > 1/(H-1)$— finds, across the 60 layer×seed combinations, a joint configuration with $\theta_{\min} \geq \theta^\star$ (worst case 84.79°). Full compliance with the regularizer lies within the joint gauge orbit: a path of exactly zero functional cost to it exists, and the inertness of section 6.2 is "no change happened despite a free path existing", not "no change was observed".

References

[1] R. A. Rankin. The closest packing of spherical caps in n dimensions. *Proceedings of the Glasgow Mathematical Association*, 2(3):139–144, 1955.

[2] V. Papyan, X. Y. Han and D. L. Donoho. Prevalence of neural collapse during the terminal phase of deep learning training. *Proceedings of the National Academy of Sciences*, 117(40):24652–24663, 2020.

[3] Z. Zhu, T. Ding, J. Zhou, X. Li, C. You, J. Sulam and Q. Qu. A geometric analysis of neural collapse with unconstrained features. *Advances in Neural Information Processing Systems* (NeurIPS), 2021.

[4] F. Pernici, M. Bruni, C. Baecchi and A. Del Bimbo. Regular polytope networks. *IEEE Transactions on Neural Networks and Learning Systems*, 2021.

[5] W. Liu, Y. Wen, Z. Yu, M. Li, B. Raj and L. Song. SphereFace: deep hypersphere embedding for face recognition. *IEEE Conference on Computer Vision and Pattern Recognition* (CVPR), 2017.

[6] H. Wang, Y. Wang, Z. Zhou, X. Ji, D. Gong, J. Zhou, Z. Li and W. Liu. CosFace: large margin cosine loss for deep face recognition. *IEEE Conference on Computer Vision and Pattern Recognition* (CVPR), 2018.

[7] J. Deng, J. Guo, N. Xue and S. Zafeiriou. ArcFace: additive angular margin loss for deep face recognition. *IEEE Conference on Computer Vision and Pattern Recognition* (CVPR), 2019.

[8] P. Michel, O. Levy and G. Neubig. Are sixteen heads really better than one? *Advances in Neural Information Processing Systems* (NeurIPS), 2019.

[9] E. Voita, D. Talbot, F. Moiseev, R. Sennrich and I. Titov. Analyzing multi-head self-attention: specialized heads do the heavy lifting, the rest can be pruned. *Annual Meeting of the Association for Computational Linguistics* (ACL), 2019.

[10] A. M. Nicolicioiu, A. L. Nicolicioiu, B. Alexe and D. Teney. Learning diverse features in vision transformers for improved generalization. *ICML 2023 Workshop on Spurious Correlations, Invariance and Stability*, 2023.

[11] A. Dosovitskiy, L. Beyer, A. Kolesnikov *et al.* An image is worth 16x16 words: Transformers for image recognition at scale. *International Conference on Learning Representations* (ICLR), 2021.

[12] A. Vaswani, N. Shazeer, N. Parmar *et al.* Attention is all you need. *Advances in Neural Information Processing Systems* (NeurIPS), 2017.

[13] X. Pan, A. Philip, Z. Xie and O. Schwartz. Dissecting query-key interaction in vision transformers. *Advances in Neural Information Processing Systems* (NeurIPS), 2024. arXiv:2405.14880.

[14] H. Yang, H. Yin, M. Shen, P. Molchanov, H. Li and J. Kautz. Global vision transformer pruning with Hessian-aware saliency. *IEEE/CVF Conference on Computer Vision and Pattern Recognition* (CVPR), 2023.

[15] M. H. Uddin, L. Seymour and S. Baidya. HEART-ViT: Hessian-guided efficient dynamic attention and token pruning in vision transformer. *arXiv preprint* arXiv:2512.20120, 2025.

[16] S. Biderman, H. Schoelkopf, Q. G. Anthony, H. Bradley, K. O'Brien, E. Hallahan, M. A. Khan, S. Purohit, U. S. Prashanth, E. Raff, A. Skowron, L. Sutawika and O. van der Wal. Pythia: a suite for analyzing large language models across training and scaling. *International Conference on Machine Learning* (ICML), 2023.

[17] A. Ahmad, A. Joshi and A. Modi. Beyond components: singular vector-based interpretability of transformer circuits. *Advances in Neural Information Processing Systems* (NeurIPS), 2025.

[18] G. Franco, C. Loughridge and M. Crovella. Singular vectors of attention heads align with features. *arXiv preprint* arXiv:2602.13524, 2026.

[19] H. Yamagiwa, Y. Takase and H. Shimodaira. Measuring affinity between attention-head weight subspaces via the projection kernel. *arXiv preprint* arXiv:2601.10266, 2026.

[20] S. Ashkboos, A. Mohtashami, M. L. Croci, B. Li, P. Cameron, M. Jaggi, D. Alistarh, T. Hoefler and J. Hensman. QuaRot: outlier-free 4-bit inference in rotated LLMs. *Advances in Neural Information Processing Systems* (NeurIPS), 2024.

[21] Z. Liu, C. Zhao, I. Fedorov, B. Soran, D. Choudhary, R. Krishnamoorthi, V. Chandra, Y. Tian and T. Blankevoort. SpinQuant: LLM quantization with learned rotations. *International Conference on Learning Representations* (ICLR), 2025.